ZOMATO

Restaurant Clustering & Sentiment Analysis

Project Report

Unsupervised Machine Learning | NLP | Clustering

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Project Type: Individual | Capstone Project

GitHub: https://github.com/Raj-Verma-1998/Zomato.git

1. Project Summary

Zomato is India's leading restaurant aggregator and food-delivery platform, operating across hundreds of cities with millions of customer reviews and thousands of restaurant listings. This project performs a complete, end-to-end data science pipeline on two Zomato datasets — restaurant metadata (name, cuisine, cost, timing, location) and customer reviews (text, star rating, reviewer metadata).

The project is structured around two core objectives. First, Sentiment Analysis: using VADER (Valence Aware Dictionary and sEntiment Reasoner), each customer review is scored and labelled as Positive, Neutral, or Negative, revealing what customers genuinely feel — independent of the star rating they gave — and identifying gaps between perceived and actual restaurant quality. Second, Restaurant Clustering: after aggregating sentiment and rating signals to the restaurant level, features such as average cost, cuisine diversity, sentiment score, and review volume are used to cluster restaurants into distinct business segments using unsupervised ML algorithms.

The output is a production-ready clustering model saved via joblib, with cluster profiles that directly map to business strategies: personalised recommendations, partner quality auditing, targeted marketing, and premium partnership identification.

2. Problem Statement & Business Objective

India's restaurant industry is one of the fastest-growing sectors, with Zomato cataloguing tens of thousands of outlets across every major city. Four core business challenges were identified:

- **Customer Experience Gap:** Star ratings alone do not capture nuanced customer sentiment — a 3-star restaurant may have overwhelmingly positive reviews about food but complaints about service.
- **Restaurant Discovery:** Users struggle to find the best restaurant for their budget and taste in a cluttered marketplace.
- **Business Intelligence:** Restaurant owners lack visibility into how their sentiment compares to competitors in the same segment.
- **Targeted Marketing:** Zomato's marketing team cannot personalise campaigns without knowing which restaurants belong to which tier.

BUSINESS OBJECTIVE

Analyse customer review sentiments using NLP and cluster restaurants into meaningful segments using unsupervised ML, enabling smarter recommendations, better partner insights, and data-driven marketing strategies.

3. Dataset Description

3.1 Restaurant Metadata

Column	Description
Name	Restaurant name (primary join key)
Links	Zomato listing URL
Cost	Average cost for two (₹) — requires cleaning (symbols, commas)
Collections	Curated collection tags — dropped (>85% missing)
Cuisines	Comma-separated cuisine types — multi-valued
Timings	Opening/closing hours — used for late-night segment flag

3.2 Customer Reviews

Column	Description
Restaurant	Restaurant name (foreign key for join)
Reviewer	Reviewer display name
Review	Free-text review body — primary NLP target
Rating	Star rating 1–5 (mixed types — some entries are strings like 'Like')
MetaData	Reviewer follower/following counts — used for critic identification
Time	Review timestamp
Pictures	Number of photos attached

KEY OBSERVATION

The Reviews dataset exhibits strong positivity bias — most customers give 4–5 stars. This makes raw ratings insufficient for quality discrimination, which is the primary motivation for sentiment analysis using NLP.

4. Data Wrangling

4.1 Steps Performed

1. Stripped whitespace from column names in both dataframes.
2. Removed exact duplicate rows from both datasets.
3. Dropped the Collections column (>85% missing, no analytical value).
4. Cleaned the Cost column using regex to strip ₹ symbols and commas, then converted to float.
5. Imputed missing Cost values using cuisine-level median — significantly more accurate than global median because North Indian and Continental restaurants have very different price distributions.
6. Coerced non-numeric Rating entries (e.g., 'Like') to NaN using pd.to_numeric with errors='coerce'.
7. Dropped reviews with null text — VADER cannot process empty strings.
8. Engineered num_cuisines by counting comma-separated values in the Cuisines field.
9. Classified restaurant timings into three categories: Regular Hours, Late Night, and 24 Hours.

4.2 Outlier Handling

IQR-based Winsorisation (3×IQR rule) was applied to the Cost column. Outliers were capped rather than removed — preserving luxury restaurants as a valid analytical data point and preventing bias toward mid-range restaurants in the clustering model. The 3×IQR rule (rather than the more aggressive 1.5×IQR) was chosen to allow a wider range of legitimate cost variation.

5. Exploratory Data Analysis

A total of 15 charts were produced during EDA. The key findings from each group are summarised below.

5.1 Cost & Cuisine Distribution (Charts 1–2)

The cost distribution is strongly right-skewed — the majority of restaurants are priced between ₹200–₹800 for two people, with a long tail extending to ₹3,000+. The median is significantly lower than the mean, confirming the presence of luxury outliers and justifying log-transformation before distance-based clustering. The top 3 cuisines by restaurant count are North Indian, Chinese, and Fast Food — consistent with pan-India preferences. Continental and Italian represent a premium niche.

5.2 Rating & Review Volume (Charts 3–4)

The rating distribution is left-skewed — most customers give 4–5 stars. This positivity bias reduces the discriminating power of raw ratings and is the primary motivation for layering in VADER sentiment analysis. Review counts follow a heavy power-law distribution — a small number of landmark restaurants receive hundreds of reviews while the majority have fewer than 20. This justified setting a minimum review threshold of 5 before including any restaurant in the clustering step.

5.3 Sentiment Results & Validation (Charts 5–6)

Over 65% of reviews were classified as Positive by VADER, with a meaningful Negative tail. Critically, Chart 6 (sentiment vs star rating) shows sentiment rising monotonically from 1-star to 5-star reviews, confirming VADER's internal consistency on this dataset and validating its use as a quality proxy.

5.4 Word Clouds & Text Insights (Charts 7–8)

Positive reviews are dominated by terms such as 'taste', 'service', 'ambiance', 'value', and dish-specific mentions — these are the primary drivers of customer delight. Negative reviews highlight 'waiting', 'cold', 'rude', 'overpriced', and 'delivery delay' — the top churning reasons. This vocabulary directly informs Zomato's automated quality alert system.

5.5 Advanced EDA (Charts 9–15)

Top-rated restaurants are niche, single-cuisine specialists rather than large chains. The cost-vs-rating scatter shows no strong linear correlation — expensive restaurants do not automatically get higher ratings. The high-cost, low-rating quadrant identifies restaurants that are a business risk. Most restaurants offer 1–3 cuisines; a significant share operate late night, representing a push-notification marketing opportunity. A small subset of reviewers have 50+ followers — food bloggers whose opinions carry disproportionate influence.

6. Hypothesis Testing

Hypothesis	Test Used	Decision
H1: High-rated restaurants have significantly higher VADER sentiment scores	Welch's two-sample t-test (unequal group sizes)	✅ Reject H_0 — $p \approx 0$
H2: Late-night restaurants have a significantly different average cost vs regular-hours restaurants	Welch's t-test	✅ Reject H_0 — late-night costs significantly higher
H3: Multi-cuisine (≥3) restaurants receive significantly more reviews than single-cuisine	Mann-Whitney U (non-parametric — review counts are right-skewed/non-normal)	✅ Reject H_0 — multi-cuisine get more reviews

Welch's t-test was chosen over Student's t-test for H1 and H2 because the two groups have significantly unequal sample sizes (high-rated vs low-rated reviews), violating the equal-variance assumption. For H3, Mann-Whitney U was used because review counts follow a heavy power-law distribution — applying a t-test would violate the normality assumption.

7. Feature Engineering & Pre-processing

7.1 NLP Text Pre-processing Pipeline (for TF-IDF)

Step	Operation	Reason
1	Expand contractions	Normalise 'don't' → 'do not' to prevent vocabulary fragmentation
2	Lowercase	Ensure case-insensitive vocabulary matching
3	Remove punctuation	Reduce noise in token vocabulary
4	Remove URLs & digit-words	Eliminate non-semantic tokens
5	Remove stopwords	Remove high-frequency, low-information words
6	Tokenise	Split text into individual token units
7	Porter Stemmer	Reduce tokens to root form for consistent vocabulary
8	TF-IDF (max 500 features, bigrams)	Down-weight common words; bigrams capture negation phrases

NOTE — VADER Applied to RAW Text

VADER sentiment analysis was intentionally run on the raw, unprocessed review text — NOT the cleaned version. VADER's scoring relies on capitalisation, exclamation marks, and negation patterns. The rephrase step was also skipped: rephrasing can destroy negation ('not good' → 'bad' breaks VADER's compound scoring).

7.2 Engineered Features

Feature	Formula / Derivation	Business Meaning
avg_sentiment	Mean VADER compound score per restaurant	Overall NLP-based quality signal
pct_positive	% reviews with compound > 0.05	Share of satisfied customers
pct_negative	% reviews with compound < -0.05	Churn risk indicator
price_per_star	Cost / avg_rating	Value efficiency metric
popularity_score	Normalised (num_reviews × avg_rating)	Combined reach and quality
value_score	avg_rating / log(Cost + 1)	Quality-to-price ratio
sentiment_gap	VADER sentiment − rating-implied sentiment	Detects hidden gems and overrated restaurants
num_cuisines	Count of comma-separated cuisine entries	Restaurant type proxy: specialist vs chain

Feature	Formula / Derivation	Business Meaning
is_open_late	1 if closes after 10 PM or 24 hours	Late-night segment flag
cuisine_* (10 flags)	Binary 1/0 for each top-10 cuisine	Domain knowledge about restaurant type

7.3 Data Scaling & Dimensionality Reduction

StandardScaler was applied to all features before clustering. KMeans and Agglomerative Clustering are distance-based algorithms — without scaling, the Cost feature (measured in hundreds of rupees) would completely dominate the binary cuisine flag features (0 or 1).

PCA was then applied retaining 95% of variance. PCA reduces the approximately 20 features (including cuisine flags) to a smaller set of uncorrelated principal components. This step removes multicollinearity (cuisine flags correlate with Cost), speeds up clustering, and enables 2D scatter visualisation of clusters via the first two principal components.

8. ML Model Implementation

8.1 Model 1 — KMeans Clustering

KMeans partitions the n restaurants into K clusters by minimising Within-Cluster Sum of Squares (WCSS/inertia). Each restaurant is assigned to the cluster whose centroid is nearest in scaled feature space. The optimal K was selected using two independent methods that both converged on K = 5: the Elbow Method (plot of inertia vs K — look for the kink) and the Silhouette Score plot (peak indicates best K).

Hyperparameter Tuning

Manual grid search over n_init ∈ {10, 20, 30} × max_iter ∈ {200, 300, 500}. Best configuration: n_init=20, max_iter=300. Increasing n_init from 10 to 20 improved silhouette score by ~0.01–0.02 by reducing the probability of convergence to a local minimum.

8.2 Model 2 — Hierarchical Agglomerative Clustering

Agglomerative clustering builds a hierarchy of clusters bottom-up: each restaurant starts as its own cluster and pairs are iteratively merged using Ward linkage, which minimises the increase in total within-cluster variance at each merge step. Ward linkage was selected after testing all four linkage methods (Ward, Complete, Average, Single) — Ward consistently produced the best silhouette score and the most compact, evenly-sized clusters. A dendrogram was visualised on a 200-restaurant subset to confirm the natural cut point for K = 5.

8.3 Model 3 — MiniBatch KMeans (on PCA features)

MiniBatch KMeans uses random subsets (mini-batches) of the data at each iteration rather than the full dataset, making it significantly faster at scale. Applied on PCA-reduced features. Hyperparameter tuning was performed over batch_size ∈ {128, 256, 512} × n_init ∈ {10, 20}. This model represents the production-deployment candidate for Zomato's full restaurant catalogue.

8.4 Model Comparison

Model	Silhouette ↑	Davies-Bouldin ↓	Calinski-Harabasz ↑	Selected?
KMeans (tuned, n_init=20)	Best	Lowest	Highest	✅ YES
Hierarchical (Ward linkage)	Good	Low	High	❌ No
MiniBatch KMeans (PCA)	Good	Moderate	Good	❌ No (production alt.)

KMeans (tuned) was selected as the final model for three reasons: highest silhouette score, stability across random seeds (n_init=20), and most importantly — interpretability. Cluster centroids can be directly read as named business segment profiles (e.g., Cluster 2 = avg cost ₹600, avg rating 4.2, 70% positive sentiment).

9. Cluster Profiles & Business Segmentation

Post-clustering, each cluster was profiled by computing the mean and median of all features per cluster. The five clusters map to the following named business segments:

Cluster	Segment Name	Key Characteristics	Recommended Zomato Action
0	Budget Local Gems	Low cost, lower review volume, strong positive sentiment from loyal regulars	New user acquisition campaigns, first-order discounts
1	Mid-Range Value	Moderate cost, high review count, balanced sentiment	Primary homepage recommendation for average users
2	Premium Justified	High cost, high rating AND high sentiment	Zomato Gold / Pro partnerships, editorial features
3	Sentiment Leaders	Moderate cost, exceptionally high positive sentiment % despite moderate rating	Hidden gem badge, organic social content, critic outreach
4	High-Volume Chains	Multi-cuisine, high review count, moderate sentiment — large brand exposure	B2B partner retention, loyalty programme anchor

Feature Importance (Clustering)

Feature importance in clustering is measured by centroid spread — the features where cluster centroids are furthest apart in scaled feature space are the most discriminating. Top 3 features: avg_sentiment, Cost (scaled), pct_positive. Cuisine flags provided secondary discrimination between specialist and chain-type clusters.

10. Business Recommendations

10. Replace raw star ratings in all ranking algorithms with a Composite Quality Score = weighted(avg_rating + VADER_sentiment). This reduces gaming and better reflects true customer experience.
11. Deploy the 5-tier cluster segmentation to power homepage personalisation: serve Budget clusters to first-time users, Premium clusters to high-spend returning users.
12. Build a Food Critic Detection Pipeline using reviewer follower counts to weight influential reviews more heavily in quality scores — flag reviewers with 50+ followers as critics.
13. Build an automated quality alert system: if a restaurant's negative-review term frequency spikes on words like 'hygiene', 'cockroach', or 'food poisoning', trigger an urgent partner quality review.
14. Create a Late-Night filter with targeted push notifications to users browsing past 9 PM — a high-conversion micro-moment marketing opportunity validated by the operating hours analysis.
15. Surface Sentiment Leader restaurants with a 'Hidden Gem' badge on listing pages — these are restaurants with positive VADER scores that are underrated by star ratings alone.

11. Technical Pipeline Summary

Stage	Tools / Techniques Used
Data Loading	pandas read_csv, encoding='utf-8', exception handling
Data Wrangling	Regex cleaning, cuisine-level median imputation, Winsorisation (3×IQR)
EDA	matplotlib, seaborn, WordCloud — 15 charts across cost, rating, sentiment, cuisine, timing, reviewer metadata
NLP Pre-processing	NLTK: contractions, lowercase, punctuation removal, stopwords, tokenisation, Porter Stemmer
Sentiment Analysis	VADER SentimentIntensityAnalyzer (compound score, 3-class label)
Text Vectorisation	TF-IDF (max_features=500, ngram_range=(1,2), min_df=3)
Feature Engineering	10 engineered features + 10 cuisine binary flags, restaurant-level aggregation
Scaling	StandardScaler (zero mean, unit variance)
Dimensionality Reduction	PCA (n_components=0.95 — retains 95% variance)
Clustering	KMeans (final), AgglomerativeClustering (Ward), MiniBatchKMeans (PCA) — K=5
Evaluation	Silhouette Score, Davies-Bouldin Index, Calinski-Harabasz Score
Hyperparameter Tuning	Manual grid search: n_init × max_iter (KMeans), linkage method (Hierarchical), batch_size (MiniBatch)
Cluster Profiling	Centroid analysis, mean/median per cluster, feature importance by centroid spread
Deployment	joblib model + scaler + PCA saved; sanity-check reload cell included

12. Conclusion

This project successfully demonstrated a complete, end-to-end unsupervised machine learning pipeline on real-world Zomato data. By combining NLP-based sentiment analysis with multi-feature restaurant clustering, the project surfaces business insights that star ratings alone cannot provide: hidden gems, overpriced underperformers, brand-anchor chains, and late-night specialists.

The final KMeans model (K=5, n_init=20) produced well-separated, interpretable clusters validated by silhouette score and confirmed by cluster profiling. The sentiment gap feature — measuring the divergence between VADER-predicted sentiment and rating-implied sentiment — proved to be one of the most valuable engineered signals for identifying mis-rated restaurants.

All deliverables are production-ready: model artifacts are serialised via joblib, the preprocessing pipeline is reproducible from raw CSVs, and cluster labels are mapped to named business segments with clear, actionable recommendations for Zomato's product, partner, and marketing teams.